

PHASE 1 PROPOSAL

CSV Data Quality Triage Agent

Team Member 1: Parthiv Godrihal(I024)
Agent, LangChain & App Integration

Team Member 2: Nilay Jain(I029)
Data Diagnostics, Evaluation & Testing

1. Selected Activity

Activity 2 – Build a Tool-Augmented/Agentic LLM Application

The application will allow a user to upload a CSV dataset and ask questions such as, “Why is this dataset causing problems during model training?” The agent will decide which diagnostic tools to call, interpret their outputs, and produce a structured data-quality report.

2. Problem Statement

Machine-learning datasets often contain issues such as missing values, duplicate rows, class imbalance, outliers, constant columns, high-cardinality identifiers, or suspicious feature relationships. Beginners may not know which checks to perform or how each issue can affect model training. The proposed system aims to create an **adaptive CSV triage agent** that investigates a dataset by selecting only the relevant diagnostic tools instead of blindly running every check, and then explains the detected issues with evidence and recommended next steps.

3. Proposed Architecture & Components

CSV Upload / User Query Input	PromptTemplate + LCEL Prompting + pipeline composition	LangChain Agent + Tools Chooses diagnostics	Pandas / sklearn Tools Deterministic checks	OutputParser + Report Structured diagnosis
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LangChain Component	Use in the Project
PromptTemplate	Defines the data-quality agent role, user objective, available tools, constraints, and instructions for investigation.
Tool / Agent	Wraps the Pandas/sklearn diagnostic functions as tools and lets the LLM decide which tool to call and in what sequence.
OutputParser	Converts the final result into structured JSON/report fields such as issue, severity, column, evidence, and recommendation.
LCEL Composition	Composes the PromptTemplate, agent/model execution, diagnostic tool outputs, and OutputParser into one end-to-end LangChain pipeline.

4. Team-Member Roles

Parthiv Godrihal(I024)	Agent, LangChain & App Integration	Build the PromptTemplate, LCEL pipeline, agent/tool-calling logic, and OutputParser; integrate the LLM with the diagnostic tools and connect the agent workflow to the Streamlit interface, including the tool-call trace and final report view.
Nilay Jain(I029)	Data Diagnostics, Evaluation & Testing	Implement the Pandas/sklearn diagnostic checks; prepare clean and deliberately corrupted CSV datasets; validate detected issues against known ground truth; perform end-to-end testing and prepare the documentation and demo material.

5. Expected Phase-1 Outcome

A working prototype in which a user uploads a CSV, the LangChain agent selectively invokes suitable diagnostic tools, and the application returns a concise structured report showing the problem, evidence, severity, and recommended action. The demo will use deliberately corrupted datasets so the correctness of the agent’s diagnosis can be evaluated clearly.